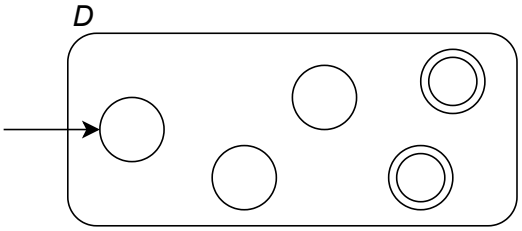
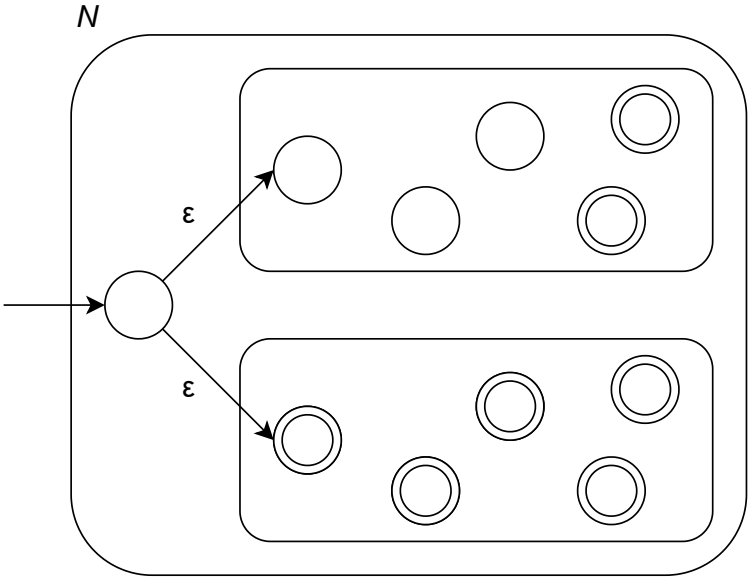

Problem 1.38. An all-NFA M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ that accepts $x \in \Sigma^*$ if every possible state that M could be in after reading input x is a state from F. Note, in contrast, that an ordinary NFA accepts a string if some state among these possible states is an accept state. Prove that all-NFAs recognize the class of regular languages.

Proof. To show that all-NFAs recognize the class of regular languages, we show that for any regular language A and a DFA D, there exists an all-NFA that recognizes A.



DFA D that recognizes some regular language A.

An all-NFA N that recognizes A can be constructed as shown in the following diagram.



An all-NFA N constructed using two copies of DFA D. The first copy (top) is exactly the same as D. The second copy (bottom) has all states marked as accept states.

□